

NBS Foundations

CUNY EES 79903

Seminar on Nature-Based Solutions

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September 8, 2026

Lecture Overview:

- 415-430 – Housekeeping: sign-ups for discussions, field trip, and projects
- 430-500 – Lecture: defining NBS, principles
- 500-615 – Discussion: Nessoover et al. (2017), Cook et al. (2025), Hall and Van Rees (2026)

Sign up for 3 dates!	Leader #1	Leader #2
Sep 8: NBS fundamentals	Grace	Jesse
Sep 15: NBS fundamentals	Michelle	Shahela
Sep 22: Structured decision making and community engagement	Jesse	Aurelia
Sep 29: Analysis – ecological	Carol	Julia
Oct 6: Analysis – co-benefits	Shahela	Aurelia
Oct 20: Analysis – engr performance	Grace	Michelle
Oct 27: Analysis – engr performance	Abdellah	Julia
Nov 3: Alternatives analysis	Aurelia	Abdellah
Nov 10: Performance-based design	Aaryan	Michelle
Nov 17: Constructability	Grace	Aaryan
Nov 24: Performance monitoring	Shahela	Jesse
Dec 1: Performance monitoring	Julia	Carol
Dec 8: Life cycle analysis	Aaryan	Abdellah
Dec 15: Course synthesis	Kyle	Brett

Narrowing options for a field trip

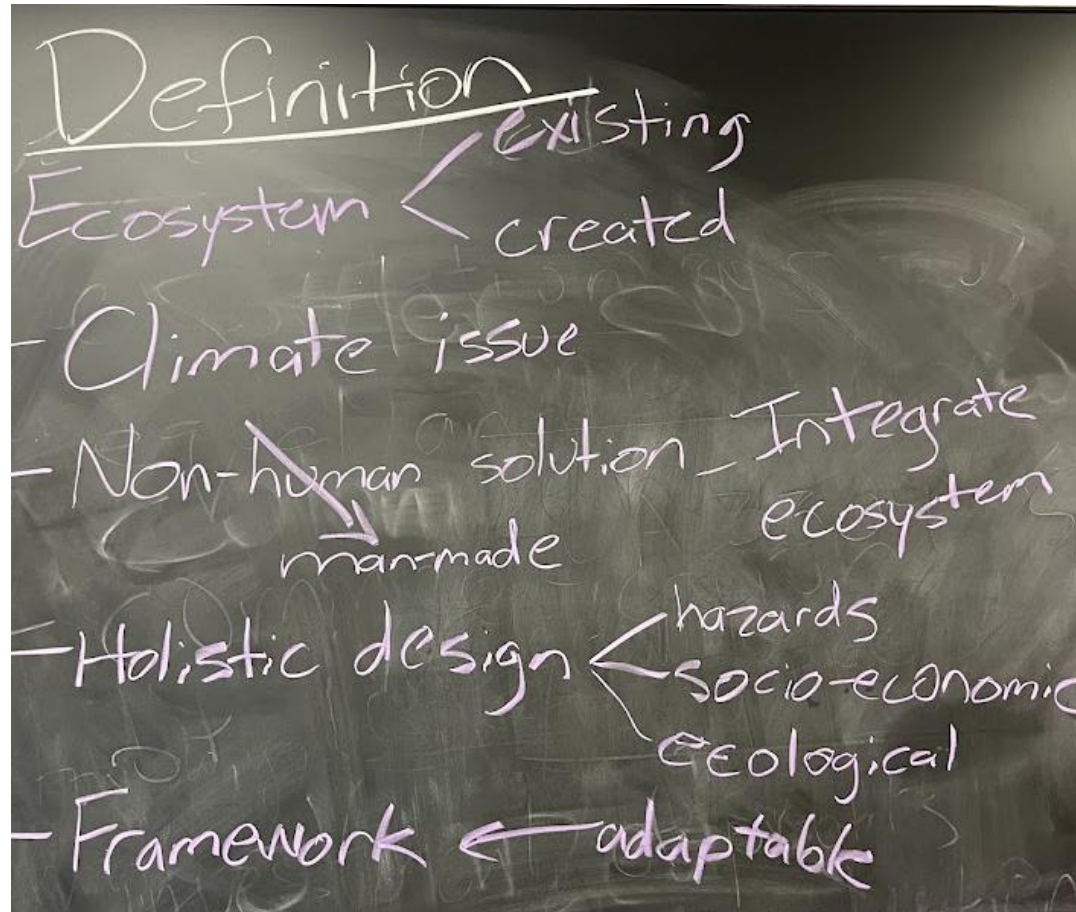
- Bad days of the week:
 - Fridays/Saturdays – field work for Grace and Michelle
 - Teaching on Friday AM – Shahela
 - Third Tuesdays – Aurelia
- Potential dates:
 - ~~Oct 2, Oct 3, Oct 10, Oct 17, Oct 18, Oct 24~~, **Oct 11 (best)**
- Potential locations:
 - Prospect Park: stormwater walk
 - Manhattan: East Side Coastal Resiliency, Battery Park, Gansevoort
 - *Jamaica Bay: Shirley Chisolm, Rockaways

Potential project topics

1. Qualitative foundations of project planning: systems context, primary and secondary design objectives, risk tolerance
2. Facilitation, engagement, and teaming: interdisciplinary dialog, ways of knowing, ladder of participation, facilitation, co-production methods
3. Ecological analysis of NBS: levels of effort, project outcomes
4. Economic analysis of NBS: levels of effort, project outcomes
5. Co-benefits analysis of NBS: levels of effort, project outcomes
6. Engineering analysis of NBS: performance, resilience, cost
7. Alternatives analysis: structured decision making, management measures, alternative development, decision support, trade-off analysis
8. Performance-based design: iterative workflows, integrated modeling, hazard resilience and reliability, adaptability
9. Constructability: material selection, regulatory requirements, developing specifications, construction oversight
10. Performance monitoring: outcomes, performance, cost
11. Project establishment: adaptive management, transition to operations
12. Asset management: condition tracking, operations, maintenance, repair, rehabilitation, decommissioning, life cycle budgeting

Foundations of nature-based solutions

How should we define nature-based solutions?





NETWORK FOR
ENGINEERING
WITH NATURE

Defining Nature-Based Solutions (in policy)

UNEP (2022): Nature-Based Solution (NbS) are “actions to **protect, conserve, restore, sustainably use and manage** natural or modified terrestrial, freshwater, coastal and marine ecosystems, which address **social, economic and environmental challenges effectively and adaptively**, while simultaneously providing human well-being, ecosystem services and resilience and biodiversity benefits”

- Slightly different definitions internationally
- Often used as umbrella term for natural infrastructure, green infrastructure, and ecosystem-based adaptation
- Typically refers to applications beyond infrastructure



Photo Credit: Cliff Cave chevrons, Mississippi River (Eddie Brauer)



NETWORK FOR
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WITH NATURE

Defining Natural Infrastructure (in practice)

(Loosely) defined by several common characteristics, not all of which may be met by any particular feature:

- Performs infrastructure services
- Consists (at least in part) of natural or living materials
- Intentionally provides environmental and social benefits beyond typical purposes
- Enhances resilience through self-adjustment



Photo Credit: Managed reconnection of the Illinois River and floodplain at the Emiquon Preserve, Illinois, USA (Photo by Doug Blodgett, The Nature Conservancy)



NETWORK FOR
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Nature-based solutions are a spectrum of green, grey, and hybrid options

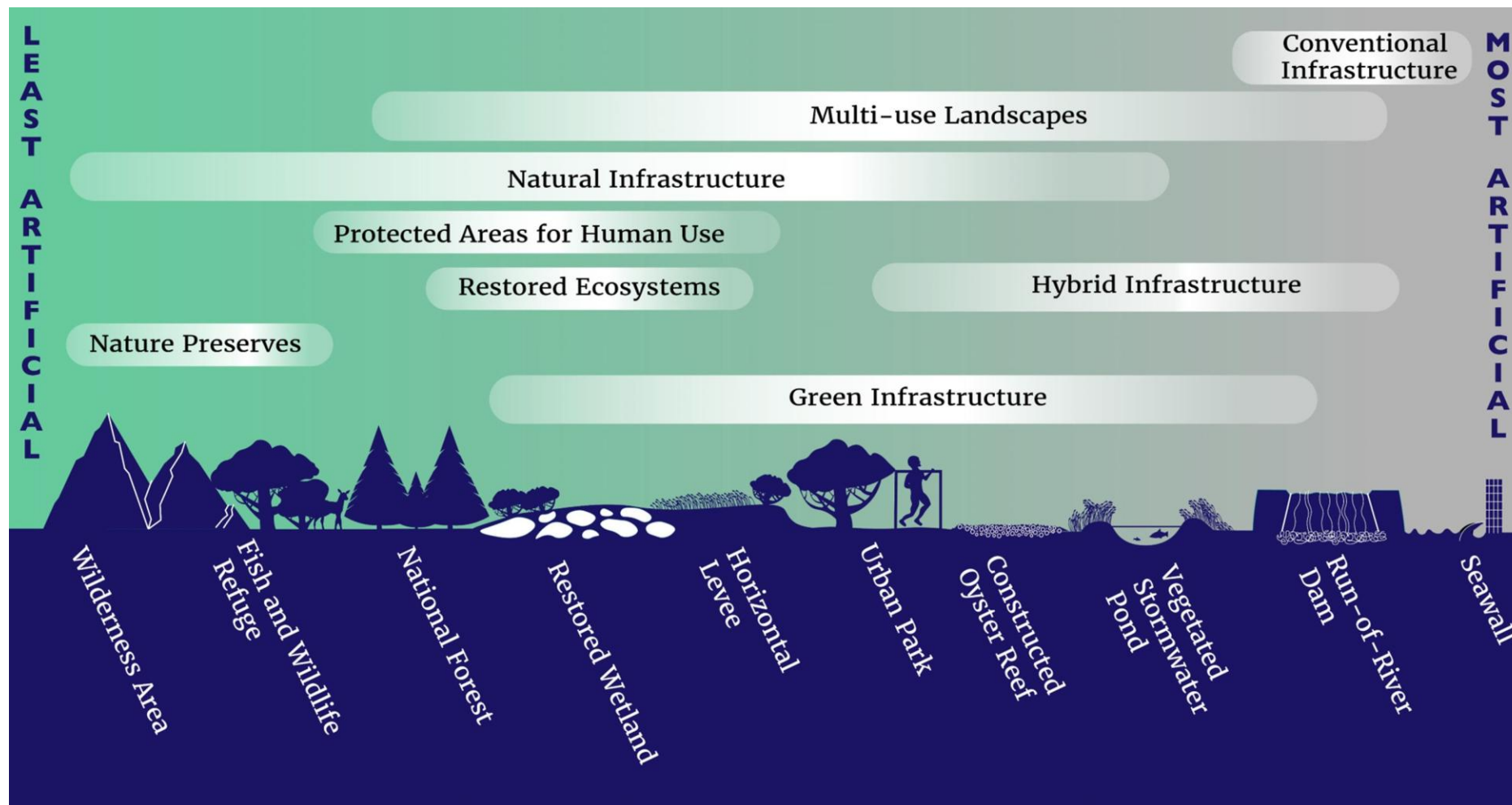
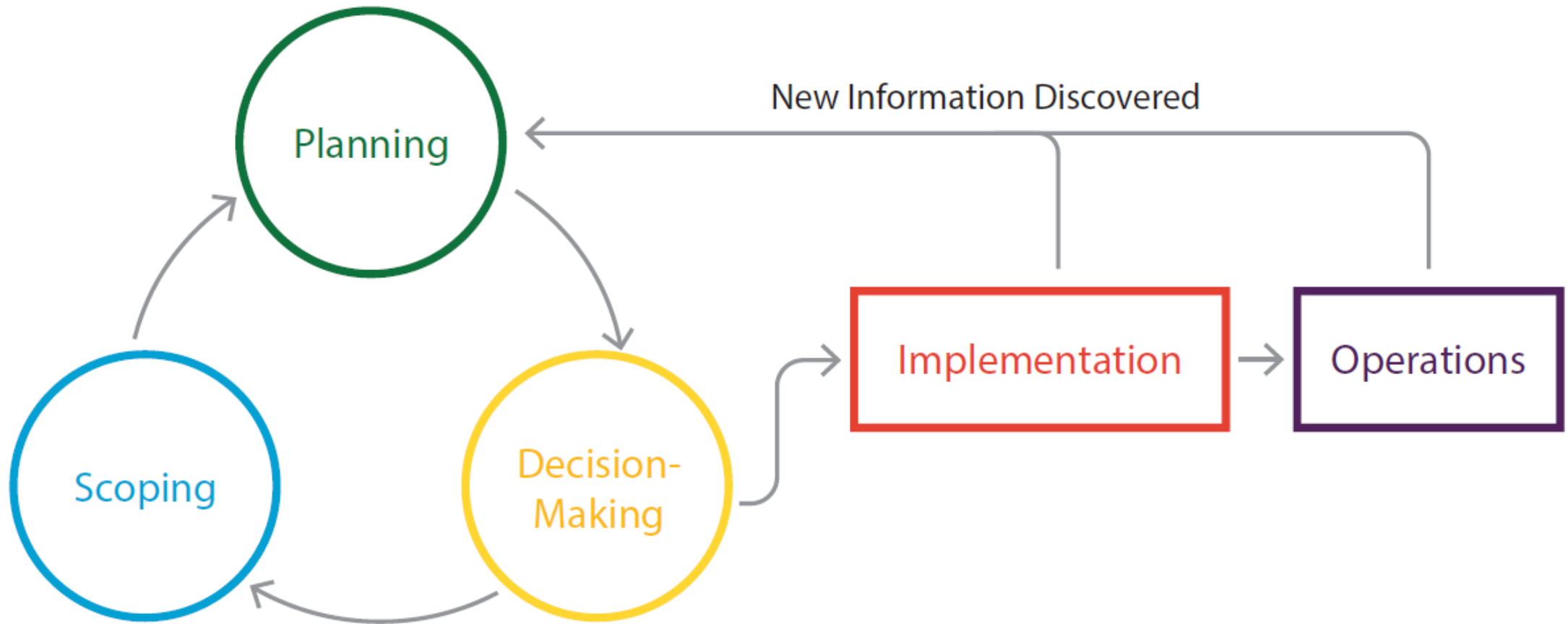
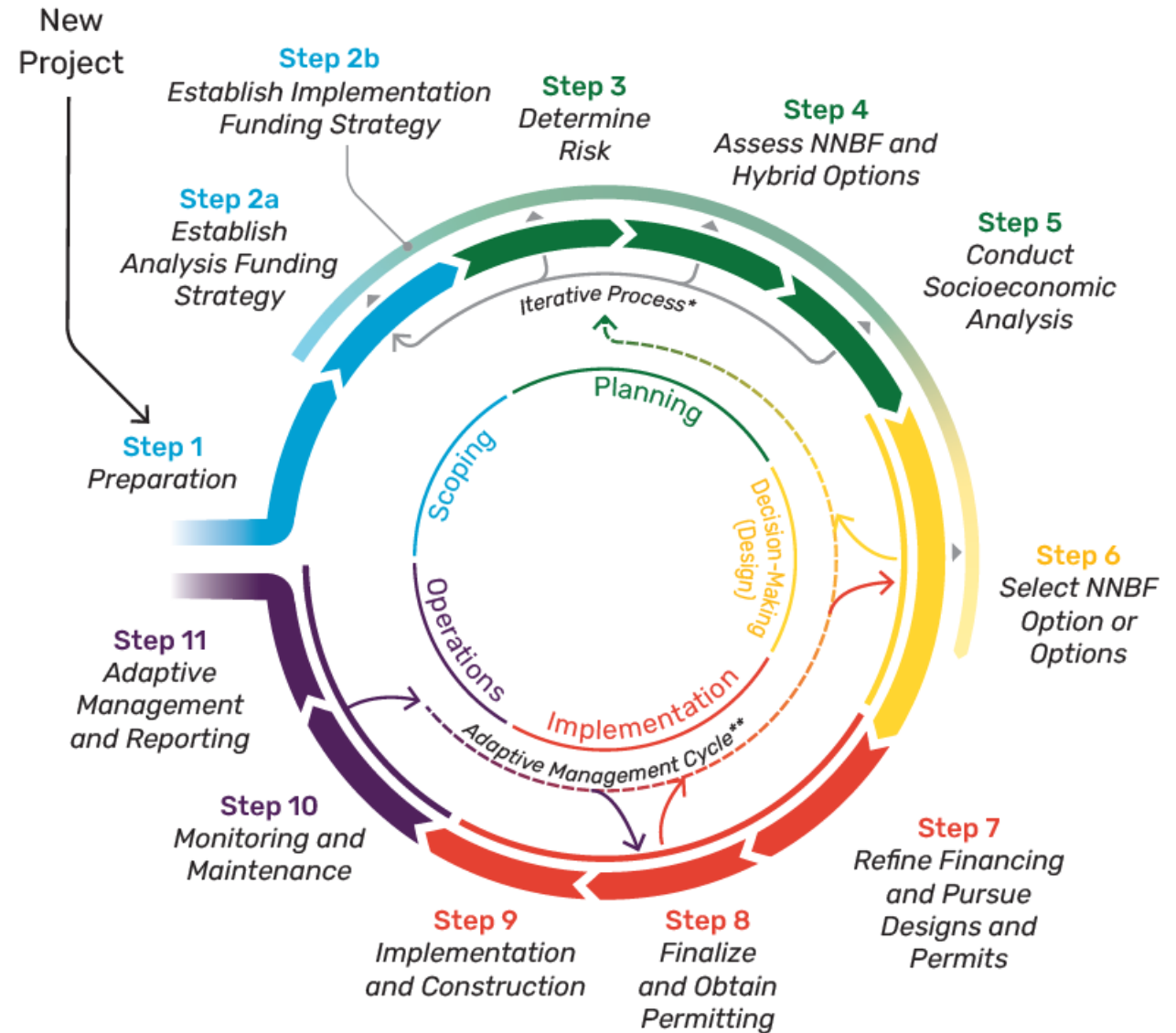


Photo Credit: van Rees et al. (2023, PNAS)

Phases of the NBS project development framework (King et al. 2021)



Detailed phases of the NBS project development framework (King et al. 2021)

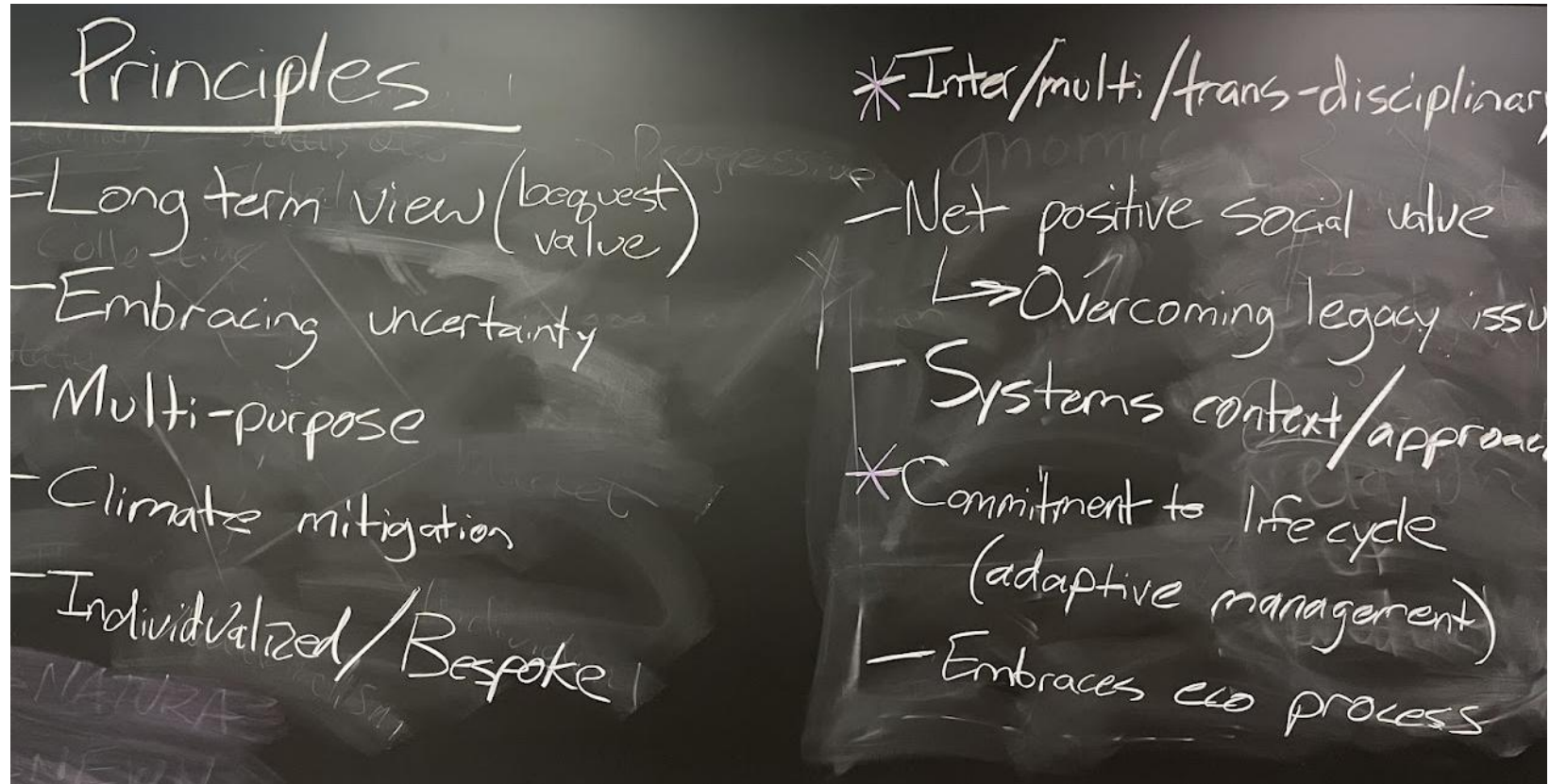


*Iterative process to identify practical and preferred NNBF options

**Adaptive management cycle that supports practical and preferred NNBF options

Before we start...

- Pair with a neighbor.
- Without revisiting the papers, list NBS principles based on your understanding of the readings.
- We will collect these as a class after 5 minutes.



NNBF principles for flood risk management (King et al. 2021)

- Use a systems approach to leverage existing components and projects and their interconnectivity.
- Engage communities, stakeholders, partners, and multidisciplinary team members to develop innovative solutions.
- Identify sustainable and resilient solutions that produce multiple benefits.
- Anticipate, evaluate, and manage risk in project or system performance.
- Expect change and manage adaptively.

Comparing and contrasting terminology

Concept	McKay's Snarky Notes
Ecological engineering & catchment systems engineering	This has been a growing discipline for a long time. Pretty much everything currently styled as NBS was ecological engineering 10 years ago.
Green/blue infrastructure	This is effectively NBS for stormwater and flooding applications.
Ecosystem approach and ecosystem-based adaptation	Seems to be mostly about effective ecological management with ancillary purposes for society and the economy.
Ecosystem services & natural capital	This is how to evaluate projects, not how to design them.

NBS Example:

Living Breakwaters

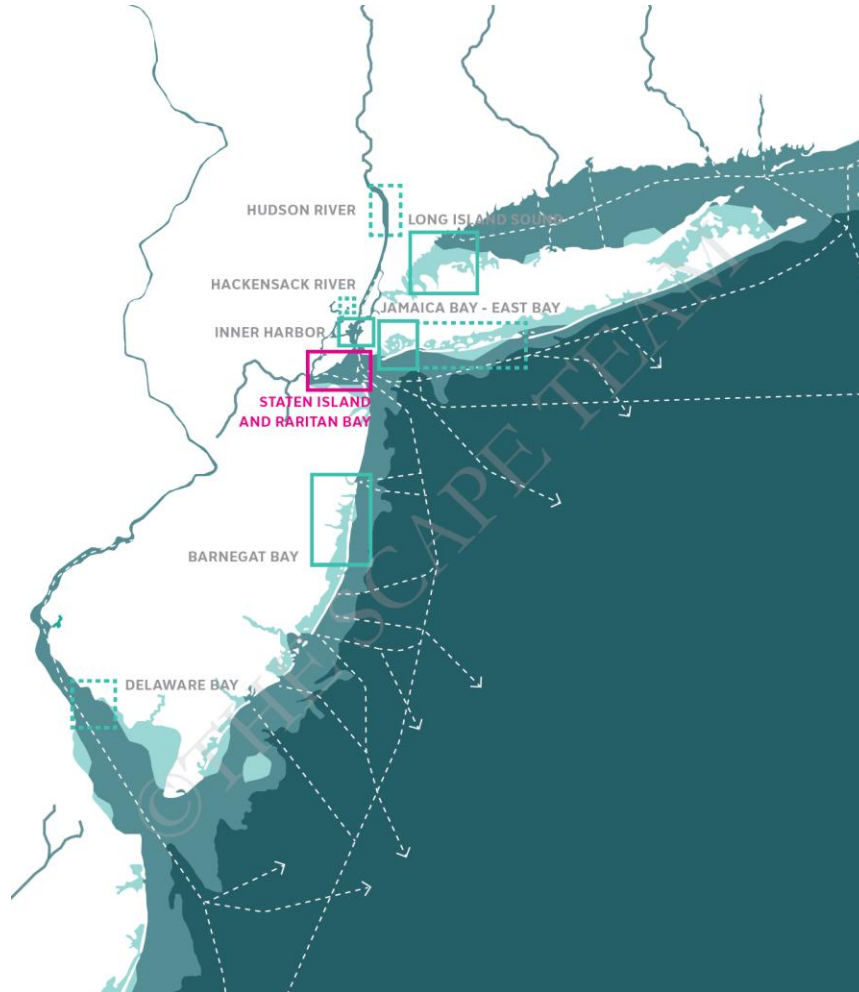
SCAPE Landscape Architecture PLLC. 2014. Living Breakwaters: IP Edition, Staten Island and Raritan Bay. Rebuild By Design. <http://www.rebuildbydesign.org/our-work/all-proposals/winning-projects/ny-living-breakwaters>. Accessed January 4, 2021.

Covered in *Good Morning, America*

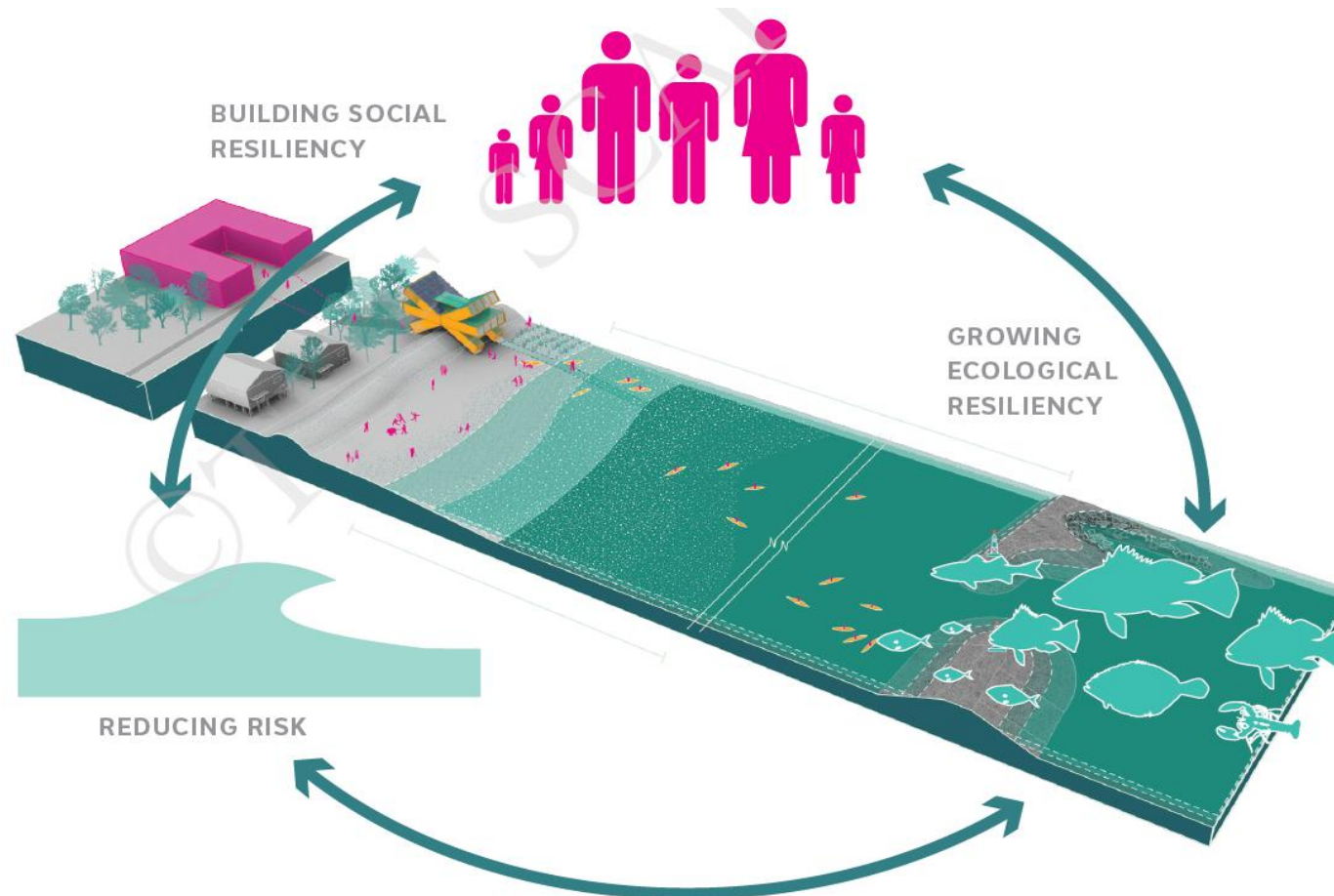
https://www.youtube.com/watch?v=XwH_YCsdLeM



Tottenville, Staten Island: Nested Flood Risk Management



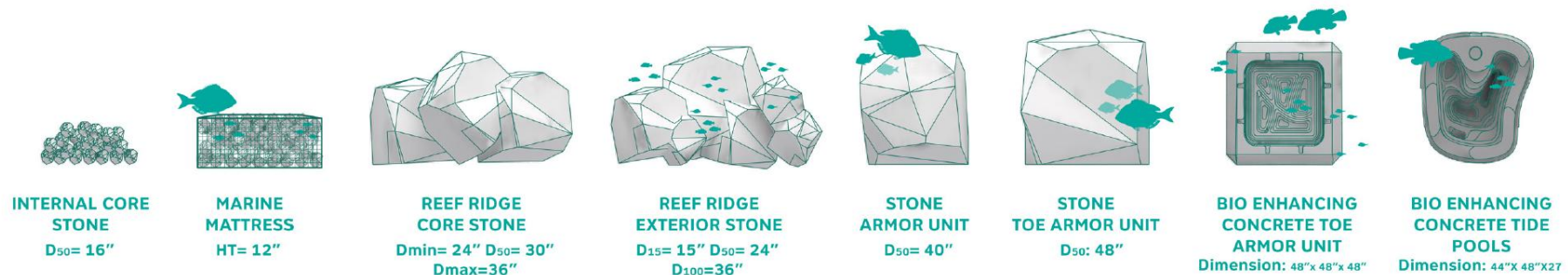
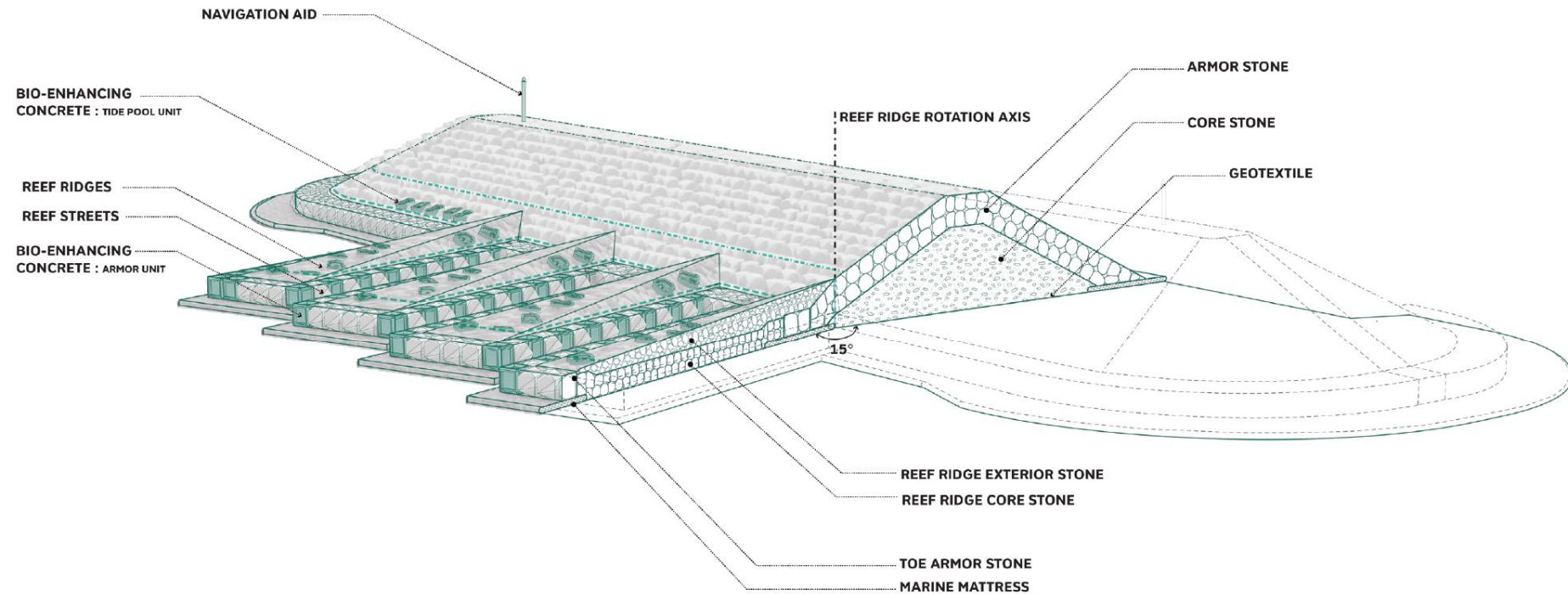
Balancing outcomes in flood risk management



Hybrid shoreline and breakwater systems



Nuanced approach to design



Paper Discussions

Nesshöver, C., Assmuth, T., Irvine, K.N., Rusch, G.M., Waylen, K.A., Delbaere, B., Haase, D., Jones-Walters, L., Keune, H., Kovacs, E. and Krauze, K., 2017. The science, policy and practice of nature-based solutions: An interdisciplinary perspective. *Science of the total environment*, 579, pp.1215-1227.

Cook, E.M., Kim, Y., Grimm, N.B., McPhearson, T., Anderson, P., Bulkeley, H., Collier, M.J., Diep, L., Morató, J. and Zhou, W., 2025. Nature-based solutions for urban sustainability. *Proceedings of the National Academy of Sciences*, 122(29), p.e2315909122.

Hall, D.M. and van Rees, C.B., 2026. Implementing nature-based solutions requires distinguishing place from space. *Ambio*, pp.1-18.

Defining NBS based on the IUCN (Nesshover et al. 2017)

- ‘the potential power of nature and the solutions it can provide to global challenges in fields such as climate change, food security, social and economic development’ (p. 1)
- ‘Healthy, diverse and well managed ecosystems lay the foundation for practical, nature-based solutions to global problems.’ (page 16)
- ‘[Nature-Based Solutions] offer unique and effective solutions to climate change.... The underlying approach is a pro-active application of the sustainable management and conservation of natural resources to address major global challenges (food security, disaster risk reduction, economy).’ (Annex I, p. 24–25)
- ‘The following principles are an initial attempt to provide some guidance on what type of interventions could (or should not) be considered as a nature-based solution.
 - The intervention delivers an effective solution to a major global challenge using nature
 - The intervention provides biodiversity benefits in terms of diverse, well-managed ecosystems
 - The intervention is cost effective relative to other solutions
 - The rationale behind the intervention can be easily and compellingly communicated
 - The intervention can be measured, verified and replicated
 - The intervention respects and reinforces communities' rights over natural resources
 - The intervention harnesses both public and private sources of funding’ (Annex 1, p. 24–25).

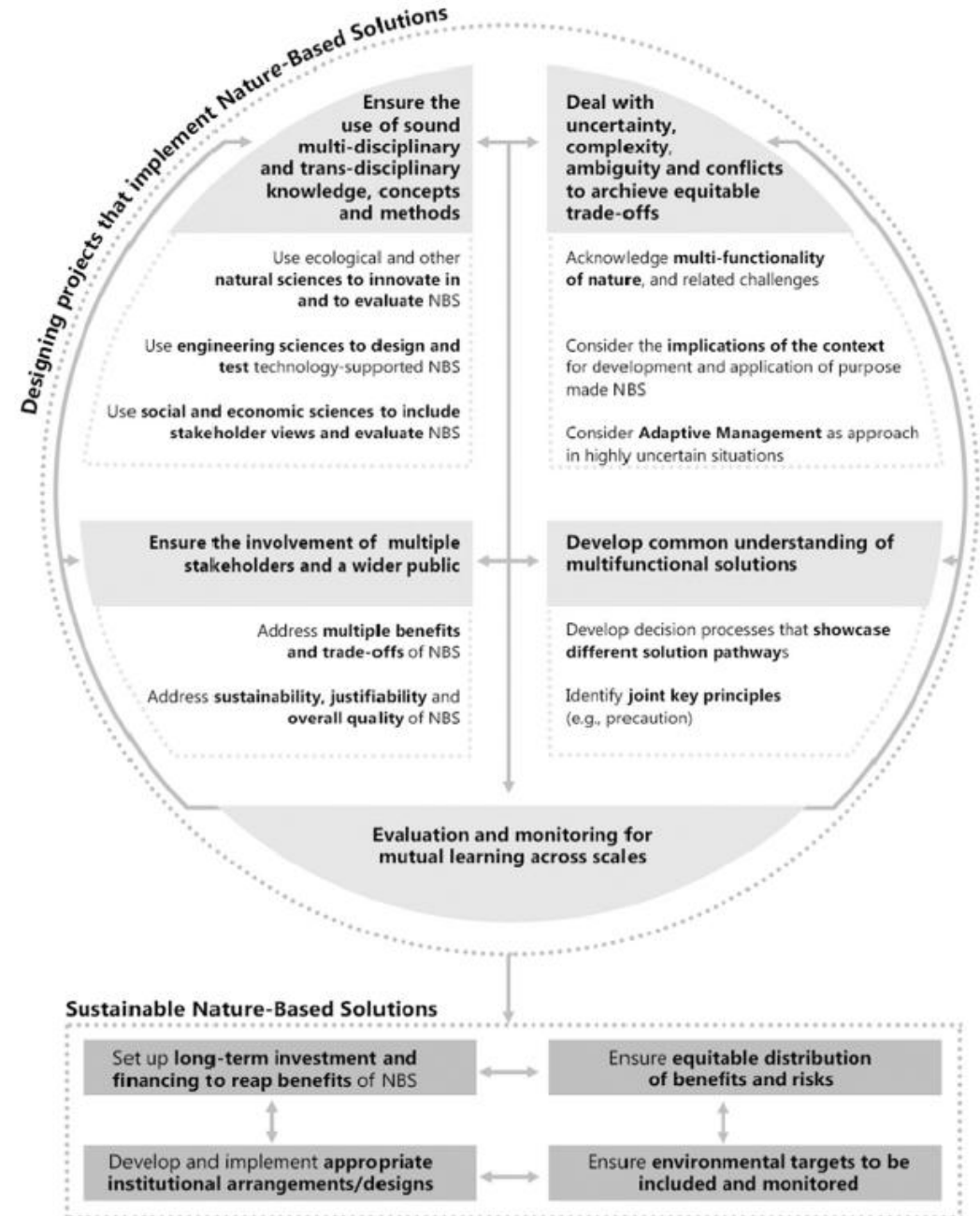
Defining NBS based on the European Union (Nesshover et al. 2017)

- ‘Nature-Based Solutions harness the power and sophistication of nature to turn environmental, social and economic challenges into innovation opportunities.’ (p. 2)
- ‘[Nature-Based Solutions] are actions inspired by, supported by or copied from nature; both using and enhancing existing solutions to challenges, as well as exploring more novel solutions, for example, mimicking how non-human organisms and communities cope with environmental extremes. Nature-Based Solutions use the features and complex system processes of nature, such as its ability to store carbon and regulate water flows...’ (p. 24)
- ‘Four principal goals have been identified that can be addressed by nature-based solutions:
 - Enhancing sustainable urbanisation...can stimulate economic growth as well as improving the environment, making cities more attractive, and enhancing human well-being.
 - Restoring degraded ecosystems... can improve the resilience of ecosystems, enabling them to deliver vital ecosystem services and also to meet other societal challenges.
 - Developing climate change adaptation and mitigation...can provide more resilient responses and enhance the storage of carbon.
 - Improving risk management and resilience...can lead to greater benefits than conventional methods and offer synergies in reducing multiple risks.’ (p. 4)

Elements for operationalizing the NBS concept (Nesshover et al. 2017)

- Dealing with uncertainty and complexity: the adaptive management approach as an example
- Ensuring involvement of multiple stakeholders
- Ensuring the sound use of multi- and trans-disciplinary knowledge
- Developing a common understanding of multifunctional solutions, trade-offs and natural adaptation
- Evaluate and monitor mutual learning

Processes and dynamics in designing and implementing NBS (Nesshover et al. 2017)



Discussion questions:

Nesshover et al. (2017)

- Who are the authors? Why did they write this paper? What do they hope it accomplishes? Does it accomplish the objectives?
- Are NBS really “solutions”?
- The IUCN definition introduces a number of concepts that do not seem unique to NBS (i.e., cost-effectiveness, communication, measurement/verification, leveraged funding). Is NBS setting the bar too high? Are these principles for NBS or just infrastructure?
- The EU definition introduces the notion of innovation. Is the goal of NBS innovation?
- Does Table 2 compare apples-to-apples?
- Does Figure 1 enhance your understanding of the paper?
- Let’s talk about ways of knowing and knowledge.

Discussion questions:

Hall and Van Rees (2026)

- Who are the authors? Why did they write this paper? What do they hope it accomplishes? Does it accomplish the objectives?
- Researchers say there is an implementation gap. Is this true?
- ASCE -> Do the right project (space). Do the project right (place).
- Balancing strategy, opportunity, and reality

Table 1 Comparison and complementarity of space-based and place-based approaches to nature-based solutions (NbS) implementation

Space-Based (Conventional) NbS Implementation Approaches	Place-Based NbS Implementation Approaches
Top-down	Bottom-up
Emphasis on present and future conditions	Emphasis on past and present, legacy, and carry-on effects
Defined by modeling and expert opinion	Defined by local knowledge and engagement
Prioritizing Alpha biodiversity	Prioritizing Beta biodiversity and endemism
Overlaying species distribution models	Collecting occurrence and detection data
Technical and academic process	Practical, implementation focus
Generalized	Contextualized and relationship-based
National and regional economic values	Local economic value, non-monetized socio-cultural values, reciprocity

Discussion questions: Cook et al. (2025)

- Who are the authors? Why did they write this paper? What do they hope it accomplishes? Does it accomplish the objectives?
- What is a network of networks? What is the return-on-investment of collaboration?
- Can ecosystems be optimized?
- Is it possible for multi-purpose projects to have “universal” metrics?



Fig. 2. Nbs networks engaged with NATURA across seven global regions.

General Discussion

- Are there “camps” of thinking in NBS?
 - NATURA vs. N-EWN
- Is NBS being held to higher standards?
 - Cook et al. (2025) -> “An identified need was for more inclusive engagement strategies...”
 - Hall and Van Rees (2026) -> “the ideal advanced in the literature is for NBS to...”
- How do we know when balance is achieved?
 - Hall and Van Rees (2026) -> “achieving parity between these viewpoints is crucial”
- Why aren’t people talking about constraints?
 - Land, cost estimates, time, bandwidth, urgency

Next Week

Cohen-Shacham, E., Andrade, A., Dalton, J., Dudley, N., Jones, M., Kumar, C., Maginnis, S., Maynard, S., Nelson, C.R., Renaud, F.G. and Welling, R., 2019. Core principles for successfully implementing and upscaling Nature-based Solutions. *Environmental Science & Policy*, 98, pp.20-29.

Singhvi, A., Luijendijk, A.P. and van Oudenhoven, A.P., 2022. The grey–green spectrum: A review of coastal protection interventions. *Journal of Environmental Management*, 311, p.114824.

Silliman, B.R., Hensel, M.J., Temmink, R.J., Paxton, A.B., Wieczynski, D.J., Gibert, J.P., Johnston, D.W., Noonan, D., Ferris, W.N., Hay, M.E. and Paalman, T., 2026. Ecosystem technology (ecotech): Harnessing natural processes to address global challenges. *Science advances*, 12(19), p.eaec5411.